

LESSON-3

Chemical Bonding ...

VSEPR Theory

(i) Valence bond Theory

(ii) Molecular Orbital Theory

i. $1D = 3.33564 \times 10^{-30} \text{ cm}$

$\mu = Q \times r$

1 Debye

$= 1 \times 10^{-18} \text{ esu cm.}$

$LP-LP > LP-BP > BP \approx BP$

Bond Length : Axial $>$ Equatorial

Strength : Equatorial $>$ Axial.

i. Dipole Moment (μ) = Charge (Q) $\times$ distance (r)

ii. $1D = 3.33564 \times 10^{-30} \text{ Cm}$ (Crossed Arrow $\rightarrow$)
 $1D = 1 \times 10^{-18} \text{ esu cm.}$ Coulomb.

iii. Average bond Enthalpy : Mean.

FORMAL CHARGE : $[\text{No. of Valence electrons}] - [\text{No. of Unshared electrons}]$
 $\downarrow$
(Charge on each atom) $- [\text{No. of Bonds around the Atom}]$

Fajan's rules

i. Cation: Covalent Nature increase with Increase in Size of Anion

ii. Anion: Covalent Nature increase with Decrease in Size of Cation

iii. Covalent Nature increase with increase in Charge on Either on Cation/Anion.

iv. Pseudo noble gas electronic Configuration

Polarisability $\propto$ Covalent $\propto$ Charge $\propto \frac{1}{\text{Cation Size}}$ $\propto$ Anion Size $\propto$ Pseudo Electronic Configuration

Geometries:

AB : Linear
 AB_2 : Bent ($\because CO_2$ is linear)
 AB_3 : Trigonal Pyramidal ($\because BF_3$ is Trigonal Planar)
 AB_4 : Tetrahedral.

No. of Electron Pairs.

2. Linear
3. Trigonal Planar
4. Tetrahedral
5. Trigonal bipyramidal
6. Octahedral.

Shapes:

Type	Bond Pairs	lone Pairs	Shape.
ABE_{32}	3	2	T-Shape
ABE_2	4	1	Bent
ABE_3	3	1	Trigonal Pyramidal (Carbocation)
ABE_{22}	2	2	Bent
ABE_4	4	1	See Saw
ABE_2	2	1	Bent : Trigonal Planar.
ABE_3	3	1	Trigonal pyramidal
ABE_{22}	2	2	Bent : Tetrahedral.
ABE_4	4	1	See Saw. : Trigonal bipyramidal.

Example.

ABE_{32}	3	2	T-Shape	} Octahedral
ABE_5	5	1	Square Pyramid	
ABE_{42}	4	2	Square Planar	

Trigonal bipyramidal

Ionisation Energy $\propto \frac{1}{r_B}$

Electron Affinity $\propto \frac{1}{r_B}$

Lattice Energy $\propto \frac{1}{r_B}$
(U)

$L \propto \frac{z^+ z^-}{r}$ (When Anion Size is Constant)
Size of Cation (Check Charge if Size is Same)

$L \propto \text{Ionic} \propto \text{MPE} \propto \text{BP}$

Applications: $IC \propto \text{MPE} \propto \text{B-P} \uparrow$

$IC \propto \text{Conductivity}$

Solubility: Hydration Energy > Lattice

$H \propto \frac{1}{\text{Size}}$

- Water has higher Dielectric constant.

Lewis Dot Structure:

$n_1 = \text{no. of v.e of atoms}$

$n_2 = (8 \times \text{no. of Atoms (Other than H)}) + 2 \times H$

$n_3 = n_2 - n_1 = \text{NO. OF Bonded } e^-$

$n_4 = n_2 - n_3$

Lone Pairs.

$F.C = \text{Valency} - \text{No. of Non Bonding electrons} - \text{No. of Bonds Electrons} \dots$

% Ionic Charge $\phi = \frac{\text{Observed } \mu}{\text{calculated } \mu} \times 100\%$

VBT:



** σ Bond $>$ π Bond
 (** ~~Side-Side~~ Head Head (Side by Side)

$P-P > S-P > S-S > \textcircled{D-P}$
 Inters nuclear. $\downarrow$ Colates at π Side by Side.

$$n = 1-1 > 1-2 > 2-2$$

Bond Strength $\propto$ Overlap.

$$\propto \frac{1}{\text{Size}} \propto \frac{1}{n} \quad (\because n \rightarrow \text{Shell Number})$$

- i. $S+S$ a
- ii. $S+P_x$ x a
- iii. $S+P_y$ y a
- iv. $S+P_z$ z a
- v. $S+P_x$ y or z no Bond

Diff-Diff:

$\rightarrow$ Any AXIS P_x+P_y $x/y/z$ no Bond.

$\rightarrow$ Same-Same

$$P_x+P_x = x \quad a$$

$\rightarrow$ Diff AXIS

$$P_x+P_y = y/z \quad \pi$$

$\rightarrow$ UnCommon AXIS

$$P_x+d_{xy} = y \quad \pi$$

$$P_y+d_{xy} = x \quad \pi$$

• For coordinate molecules check CFST. (Cristal field theory)

Same Same
Same AXIS

Bond Order :

- i. $d_{xy} + d_{xy} \cdot xy \pi$
- ii. $d_{xy} + d_{xy} \cdot z \delta \text{ bond}$
- iii. $d_{x^2-y^2} + d_{x^2-y^2} \cdot z \delta \text{ bond}$

$$1 \rightarrow 0.5$$

$$2 \rightarrow 1$$

$$3 \rightarrow 0.5$$

$$4 \rightarrow 0$$

$$5 \rightarrow 0.5$$

$$6 \rightarrow 1$$

$$7 \rightarrow 0.5$$

$$8 \rightarrow 0$$

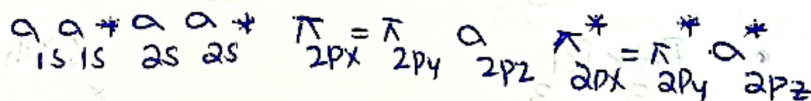
Hybridisation :

1. Ve^-
2. bonds
3. If Any Oxygen Present give $2e^-$
4. L.P $\rightarrow$ Consider it as Super Pair

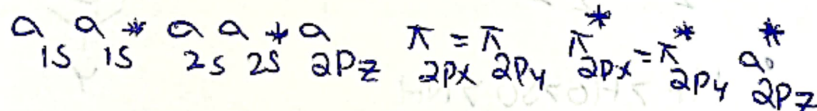
Average Bond Order / Fractional Bond :

MOT :

- i. UP to $14e^-$ (2s-2p Mixing) $xyz \cdot xyz$



- ii. Above $14e^-$: (2s-2p NOT there)



Odd No. of electrons $\rightarrow$ Para

Even No. of electrons $\rightarrow$ Diamagnetic

(Exception O_2)

- if 2s2p is not there then B_2 is Also Diamagnetic...

Bond Dissociation Energy $\propto B.O.$

$$H = \frac{1}{2} \left[\begin{array}{l} \text{NO. of } e^- \text{ in Valence} \\ + \\ \text{Hybridisation} \quad \text{NO. of Monovalent atoms} \\ + \\ \text{Charge on Anion } (-) \\ + \\ \text{Charge on Cation } (+) \end{array} \right]$$

$$8 \rightarrow 0$$

$$9 \rightarrow 0.5$$

$$10 \rightarrow 1$$

$$11 \rightarrow 1.5$$

$$12 \rightarrow 2$$

$$13 \rightarrow 2.5$$

$$14 \rightarrow 3$$

$$15 \rightarrow 3.5$$

$$16 \rightarrow 4$$

$$17 \rightarrow 4.5$$

$$18 \rightarrow 5$$

$$19 \rightarrow 5.5$$

$$20 \rightarrow 0$$

Example: $O_2^{2-}: 18e^-$: $\uparrow s^2 \uparrow p^5 \uparrow s^2 \uparrow p^5 \uparrow s^2 \uparrow p^2 \uparrow p^2 \uparrow p^2 \uparrow p^2$

$$B.O = \frac{1}{2} [B.M.O - A.B.M.O]$$

$$B.O = \frac{1}{2} [B.M.O - A.B.M.O]$$

$$= \frac{1}{2} [10 - 8] = \frac{2}{2} = 1$$

$l \neq 0$ Polar
 $l = 0$ Nonpolar
 $l \propto EN \propto \frac{1}{\text{Bond Angle}}$
 Bent or Angular have l .

Dipole Movement.

lone pair l .

• $HF > HO > NH > NF$ (Check by NO. of Lone Pairs)

• $CHCl > CHF > CHBr > CHI$

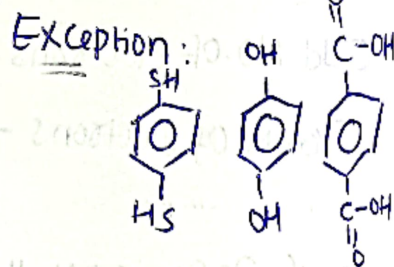
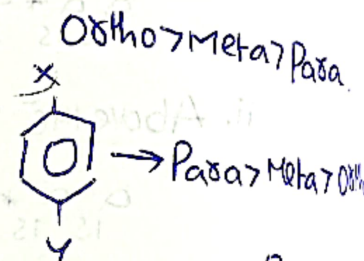
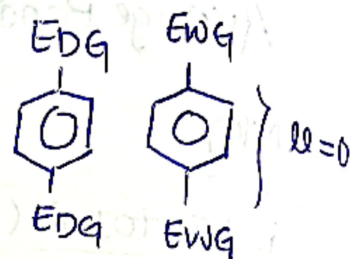
• $NO_2^- > NO_2 > NO_2^+$

• $BF_3 < NF_3$ and NH_3

• $PH < PCl$

• $HO < HO$

• $HF > HO > SO > NH$



Bond Parameters:

B.L $\propto$ No. of Shells $\propto \frac{1}{EN} \propto \frac{1}{B.O.} \propto \frac{1}{\text{Resonance}}$

$$d_{AB} = \delta_A^+ + \delta_B^- \text{ (Non Polar)}$$

$$d_{AB} = \delta_A + \delta_B - 0.09 \Delta \chi$$

$BA \propto \text{Character} \propto \frac{1}{L.P} \propto C.A.E.N \propto \frac{1}{S.A.E.N}$
 Central Atom
 Electron
 Negativity
 (Side Atom)

Drago's: From 3rd Period Hydrogen of N & O family or higher

Bond Energy:

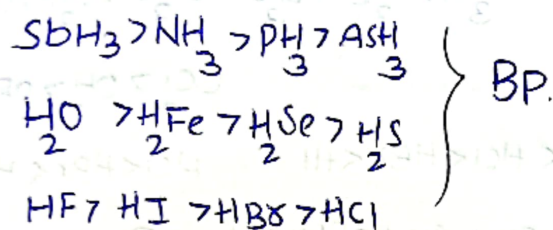
- EN of 2.5 or less
- At least one lone pair.

$$BE \propto \frac{1}{BL} \propto B.O \propto \frac{1}{\text{Lone Pairs}} \propto EN \propto \text{Character} \propto \frac{1}{\text{Bond Size}} \propto \text{Polarity}$$

Hydrogen Bonding $\propto$ Viscosity

- Intra $\propto$ B.P
- Inter $\propto$ Solubility

(Intra $\times$ Inter) Bond Strength.



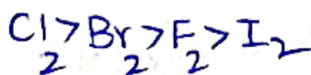
- HCl (Doesn't form Hydrogen bond)
- Hydrogen bond forms only for compounds in which F or O or N atoms are attached to hydrogen atom.
- Compounds having Direct Attachment of Any of F or O or N only will form hydrogen bond.

Donating: NH_2

Withdrawing: NO_2

Questions related to Dipole Moment.

- All halogens/Hydrogen is Monovalent.
- Bond Dissociation Energy



- Bonding Electrons More $\rightarrow$ More Stable
- Anti Bonding e^- More $\rightarrow$ Less Stable